

EXP 7

Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits. Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits.

AIM

To develop an Online Quiz Application using Java Servlets that stores user preferences (such as quiz category or difficulty level) using cookies and retrieves them on subsequent visits.

ALGORITHM

1. Start the application

User opens the quiz application.

2. Accept User Preferences

User selects:

Quiz category (e.g., Java, HTML)

Difficulty level (Easy/Hard)

3. Create Cookies

Create cookies for:

Category

Difficulty

4. Set Cookie Expiry

Set cookie lifetime using `setMaxAge()`.

5. Store Cookies

Send cookies to the client browser.

6. On Subsequent Visits

Retrieve cookies using `request.getCookies()`.

7. Check Stored Preferences

If cookies exist:

Extract stored values.

8. Display Preferences

Show:

Selected category

Difficulty level

Display quiz based on preferences.

9. Handle Missing Cookies

If no cookies:

Show default quiz or ask user to select preferences.

10. Stop the application

STEPS:

Create UI to select quiz preferences.

Create Servlet to store preferences using cookies.

Set expiry time for cookies.

Send cookies to browser.

Create another Servlet to read cookies.

Display stored preferences.

CODE :

Index.html (Quiz Preference Page)

```
<!DOCTYPE html>
<html>
<body style="font-family:Segoe UI;text-align:center;margin-top:100px;">

<h2>Select Quiz Preferences</h2>

<form action="PreferenceServlet" method="post">

    Category:
    <select name="category">
        <option>Java</option>
        <option>HTML</option>
    </select><br><br>

    Difficulty:
    <select name="difficulty">
        <option>Easy</option>
        <option>Hard</option>
    </select><br><br>

    <input type="submit" value="Start Quiz">

</form>

</body>
</html>
```

PreferenceServlet.java (Store Cookies)

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class PreferenceServlet extends HttpServlet {

    public void doPost(HttpServletRequest req, HttpServletResponse res)
        throws IOException {

        String category = req.getParameter("category");
        String difficulty = req.getParameter("difficulty");

        Cookie c1 = new Cookie("category", category);
        Cookie c2 = new Cookie("difficulty", difficulty);

        c1.setMaxAge(86400); // 1 day
        c2.setMaxAge(86400);
```

```

res.addCookie(c1);
res.addCookie(c2);

res.setContentType("text/html");
PrintWriter out = res.getWriter();

out.println("<h3>Preferences Saved!</h3>");
out.println("<a href='QuizServlet'>Start Quiz</a>");
}
}

```

QuizServlet.java (Retrieve Cookies & Display Quiz)

```

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class QuizServlet extends HttpServlet {

    public void doGet(HttpServletRequest req, HttpServletResponse res)
        throws IOException {

        String category = "Java";
        String difficulty = "Easy";

        Cookie[] cookies = req.getCookies();

        if (cookies != null) {
            for (Cookie c : cookies) {
                if (c.getName().equals("category"))
                    category = c.getValue();
                if (c.getName().equals("difficulty"))
                    difficulty = c.getValue();
            }
        }

        res.setContentType("text/html");
        PrintWriter out = res.getWriter();

        out.println("<html><body style='font-family: Segoe UI;'>");
        out.println("<h2>Online Quiz</h2>");

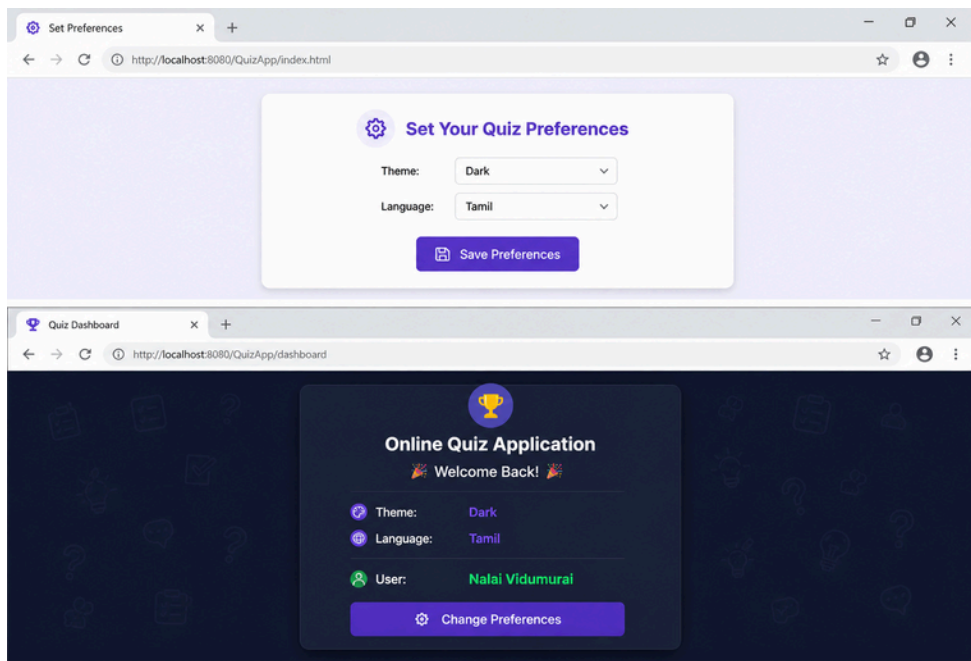
        out.println("<p><b>Category:</b> " + category + "</p>");
        out.println("<p><b>Difficulty:</b> " + difficulty + "</p>");

        // Sample Question
        if (category.equals("Java")) {
            out.println("<p>Java is a?</p>");
            out.println("a) Language b) OS");
        } else {
            out.println("<p>HTML stands for?</p>");
            out.println("a) Markup b) Machine");
        }

        out.println("<br><a href='index.html'>Change Preferences</a>");
        out.println("</body></html>");
    }
}

```

OUTPUT :



RESULT :

An Online Quiz Application is successfully developed using Java Servlets, demonstrating the use of cookies to store and retrieve user preferences, providing a personalized quiz experience on subsequent visits.

OUTPUT:

RESULT

The Servlet application successfully stores user preferences using cookies and retrieves them on subsequent visits, providing a personalized user experience.